MILITARY SPECIFICATION

FITTINGS, TUBE, FLUID SYSTEMS, SEPARABLE, DYNAMIC BEAM SEAL, GENERAL REQUIREMENTS FOR

1. SCOPE

1.1 Scope. This specification covers the basic test and design requirements for separable dynamic beam seal fittings including boss or port seal connectors for use in aerospace fluid systems. Definitions of fitting and related terms are included in paragraph 6.5.

1.2 Classification. Fittings shall be of the following types and system pressures.

Type II $-65^{\circ}F$ to $+275^{\circ}F$ Temperature Range
Type III $-65^{\circ}F$ to $+390^{\circ}F$ Temperature Range
Type IV $-65^{\circ}F$ to $+600^{\circ}F$ Temperature Range

System Pressure	Where the cutout pressure at the main pressure controlling device is
3000 psi	3000 psig ±15%
4000 psi	4000 psig ±15%

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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SPECIFICATIONS

Military

MIL-C-5501	Caps and Plugs, Protective, Dust and Moisture Seal
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-H-6088	Heat Treatment of Aluminum Alloys

STANDARDS

Federal

FED-STD-151	Metal; Test Methods

Military

DOD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedure and Tables for Inspection by Attribute
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-280	Definitions of Item Levels, Item Exchangeability, Models, and Related Terms
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American National Standards Institute

B46.1	Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018).

Society of Automotive Engineers

ARP 603	Impulse Testing of Hydraulic Hose Assemblies, Tubing, and Fittings
ARP 1185	Flexure Testing of Hydraulic Tubing Joints and Fittings
AIR 1377	Fire Test Equipment for Flexible Hose and Tube Assemblies
AMS 4928	Bars and Forgings - 6Al 4V, Annealed - 120,000 psi Yield
AMS 4944	Tubing, Seamless, Hydraulic - 3.0Al 2.5V, Cold Worked, Stress Relieved
AMS 4965	Bars, Forgings, and Rings - 6Al 4V, Sol. & Precip. Heat Treated
AMS 5561	Tubing, Welded and Drawn - 9.0Mn 20Cr 6.5Ni 0.28N, High-Pressure Hydraulic
AMS 5643	Steel Bars, Forgings, Tubing, and Rings, Corrosion Resistant
AMS 5644	Bars and Forgings - 17Cr 7Ni 1Al
AMS 5659	Bars, Forgings and Rings, 15CR 5Ni 0.30 (Cb+Ta) 4CU, Con. Elect. Melted

(Copies of SAE publications may be obtained from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15098.)

(When requesting applicable documents, refer to both title and number. Copies of unclassified documents may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Requests for copies of classified documents should be addressed to the Naval Publications and Forms Center, via the cognizant Government representative.)

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3. REQUIREMENTS

3.1 First article. Each fitting or component furnished under this specification shall be a product containing separable and/or attachment end configurations identical to that which has been inspected and passed the first article inspection specified herein (see 4.3). The procuring activity shall be responsible for granting first article approval based on MIL-F-85421 and other documents prepared by the original contractor. The contractor-approved first article test in compliance with this specification for the specific fitting and tube attachment method used on a specific aircraft shall be acceptable for use on other aircraft without additional testing, providing the tubing material and wall thickness as well as the operating temperature and pressure are the same or lower. Above is subject to the approval of the alternate aircraft manufacturer.

3.2 Classification of requirements. The requirements for the product are classified herein as follows:

Requirements	Paragraph
Material	3.3
Design and construction	3.4
Performance	3.5
Part numbering	3.6
Dimensions	3.7
Weight	3.8
Identification of product	3.9
Workmanship	3.10

3.3 Material.

3.3.1 Fittings. The material used shall be titanium alloys or corrosion resistant steels as specified in Table I. Aluminum alloys may be used for certain Type II applications. The material shall be heat treated, as required, and have suitable surface protection to meet the test requirements of this specification. Reclaimed materials shall be utilized to the maximum extent possible providing the materials used conform to the applicable material specifications.

3.3.2 Tubing. The tubing used for the fabrication of the test assemblies shall be in accordance with Table I.

3.4 Design and construction.

3.4.1 Styles and dimensions. The styles and dimensions shall conform to the design requirements specified herein and applicable drawings as approved by the procuring activity.

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TABLE I. Fitting and tubing for test assemblies and flexure test levels.

Fitting Size	3000 PSI						4000 PSI	
	Ti Fitting Ti Tubing		Ti Fitting CRES Tubing		CRES Fitting CRES Tubing		Ti Fitting Ti Tubing	
	Tubing Wall	Stress (psi)	Tubing Wall	Stress (psi)	Tubing Wall	Stress (psi)	Tubing Wall	Stress (psi)
03	.020	20,000					.020	20,000
04	.020	20,000	.020	40,000	.020	30,000	.020	19,000
05	.020	20,000			.020	28,000	.028	18,000
06	.020	19,000	.020	30,000	.020	26,000	.028	18,000
08	.026	18,000	.026	27,500	.026	20,000	.035	17,000
10	.032	17,000	.033	24,500	.033	18,000	.044	17,000
12	.039	16,000	.039	23,000	.039	16,000	.052	17,000
16	.051	15,000	.052	22,000	.052	15,000	.070	13,000
20	.065	14,000	.032*	20,000	.054*	15,000	-	-
24	.032*	13,000					-	-

Notes:

1. All wall thicknesses expressed in inches.
2. Stress specified is the minimum bending stress for the Flexure Test per 4.6.6.
3. Titanium fittings shall be fabricated of Titanium 6Al-4V per AMS 4965.
4. CRES fittings shall be fabricated of CRES 17-4PH per AMS 5643 - Condition H1075 or CRES 15-5PH per AMS 5659 - Condition H1075.
5. Titanium tubing shall be Titanium 3Al-2.5V per AMS 4944 or AMS 4928.
6. CRES tubing shall be CRES 21-6-9 per AMS 5561.
7. For test purposes, tubing diameters and wall thicknesses shall be per table I. Fittings can be used for other tubing wall thicknesses.

* 500 psig systems.

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3.4.2 Drill offset. On fittings where the fluid passage is drilled from each end, the offset between the drilled holes at the meeting point of the drills shall not exceed .015 inch. Angular misalignment shall not exceed two degrees.

3.4.3 Reduction in fluid passage (bore) cross-section. It shall be possible to pass through the passage a ball whose diameter is 0.020 less than the minimum diameter specified for the passage (bore). On angle joints or reducers, the cross-sectional area at the junction of the fluid passages shall in no case be smaller than the cross-sectional area of the smallest passage.

3.4.4 Tube attachment. The method of joining the tubing to the end fitting (brazing, welding, mechanical attachment, swaging, etc.) shall not adversely affect the properties, strength or geometry of the tubing or end fitting to the extent that performance would be impaired. The joint shall be of a quality compatible with the design and performance requirements of this specification, and shall be inspectable by direct measurement, x-ray, or other such means. The method of tubing attachment of the test assemblies shall be as specified or approved by the procuring activity.

3.4.5 Installation tooling. Size of tool envelope required to install fittings on tubing in the vehicle system, to effect repair of the system, or to replace fittings within the system shall be kept to a minimum. The wrench flats or pads of the fittings shall permit use of standard wrenches.

3.4.6 Heat generation. Fittings attached by thermal process shall be attached by tooling designed to permit installation in vehicles without heat damage to adjacent structure or components.

3.4.7 Connector-boss interface. Connectors shall be designed to prevent extrusion or mechanical failure of seals during installation and throughout the required temperature range. Type III and IV fitting materials should be selected to maintain approximately the same coefficient of thermal expansion or contraction between the connectors and the mating boss or port, unless the seal or boss assembly design allows the use of such adverse property materials. Protection of dissimilar metals against galvanic corrosion should also be considered.

3.4.8 Selection of specifications and standards. Specifications and standards for necessary commodities and services shall be selected in accordance with MIL-STD-143.

3.5 Performance. The fitting assemblies shall meet the following performance requirements.

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3.5.1 Proof pressure. The fitting assembly shall withstand pressure equal to twice the nominal pressure of the system for five minutes without leakage, evidence of permanent deformation, or other malfunction that would affect assembly or disassembly using the specified range of torque values when tested in accordance with 4.6.2.

3.5.2 Burst pressure. The fitting assembly shall not rupture or show evidence of leakage at any pressure less than four times the nominal operating pressure when tested in accordance with 4.6.3. Tubing expansion is permissible. The fitting assembly need not meet any disassembly or assembly requirements after this test. It is not essential that the assembly actually be burst but it must be able to maintain four times the nominal system operating pressure for a minimum of 5 minutes.

3.5.3 Thermal shock. The fitting assembly shall not leak or burst during temperature and pressure cycling from the minimum system temperature to the maximum system temperature when tested in accordance with 4.6.4.

3.5.4 Impulse. The fitting assembly shall withstand 200,000 impulse pressure cycles without leakage from the fitting or the fitting-tube junction when tested in accordance with 4.6.5.

3.5.5 Flexure. The fitting assembly shall not exhibit leakage from the fitting or the fitting-tube junction prior to the completion of 10 million cycles at the test stress level specified when tested in accordance with 4.6.6.

3.5.6 Tensile strength. The fitting assembly shall withstand a tensile load equal to or greater than the burst pressure times the area of tubing based on nominal O.D., without any movement at the fitting-tube junction when tested in accordance with 4.6.7.

3.5.7 Repeated connection. The fitting assembly shall withstand 25 repeated connections at the minimum and maximum torque values specified in Table IV when tested in accordance with 4.6.8, without:

a. Leakage at any of the pressure tests of 4.6.8.

b. Inability to assembly the fitting to the point of interface by hand.

c. Nut deformation preventing engagement of nut hex with open end wrench.

3.5.8 Stress corrosion. The fitting assembly shall withstand 50 hours of salt spray exposure when tested in accordance with 4.6.9 without:

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a. Indication of cracking or pitting of the exposed surfaces of fittings and tubing within one diameter of end fitting greater than that exhibited on the balance of the tubing when visually examined with 10 power magnification.

b. Leakage or burst at a value less than four times the nominal operating pressure.

c. Indications of interior transgranular corrosive attack during metallurgical examination of longitudinal and transverse sections of the fitting and fitting-tube junction.

3.5.9 Pneumatic leakage. The fitting assembly shall withstand pneumatic pressure equal to the proof pressure for five minutes at room temperature without any visible bubbles starting after one minute at pressure or other malfunction that would affect assembly or disassembly using the range of torque values specified when tested in accordance with 4.6.10.

3.5.10 Fire. The fitting assembly shall withstand a 2000°F (1093°C), 4500 BTU/HR. (1320 watts) flame for 15 minutes when tested in accordance with 4.6.11. Heat input shall be measured by a 1/2-inch (12.7-mm) copper tube per AIR 1377. There shall be no leakage detected by visual observation or failure of the test assembly prior to the specified time.

3.6 Part numbering of interchangeable parts. All parts having the same design activity FSCM (Federal Supply Code for Manufacturers) and part number shall be interchangeable as defined in MIL-STD-280. The item identification and part number requirements of DOD-STD-100 shall govern the design activity part numbers and changes thereto.

3.7 Dimensions and tolerances. All pertinent dimensions, threads, and tolerances, where interchangeability, operation or performance of the fitting may be affected, shall be as specified on detail drawings.

3.8 Weight. Fitting weights shall not exceed those shown in Table II.

3.9 Identification of product. All fittings shall be identified in accordance with the following instructions. Marking shall be impressed, embossed, or electro-chemical etched in a location and in a manner not detrimental to the fitting. Where practical, marking shall remain visible after the fitting is installed.

3.9.1 All end fittings, space permitting, (see 6.5.2) shall be marked with the manufacturer's name or trademark, an appropriate material identification code and the dash size (mating tube) of the fitting, as a minimum.

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TABLE II. Maximum fitting weights (in pounds).

Material	CRES		Titanium						
Description	End fitting		End fitting		Connectors				
	Internally threaded	Externally threaded	Internally threaded	Externally threaded	Union	Elbow 45°	Elbow 90°	Tee	Cross
Size									
03	.016	.016	.014	.014	.015	.022	.015	.017	.019
04	.024	.024	.020	.020	.016	.024	.020	.025	.030
05	.026	.029	.024	.027	.019	.030	.031	.036	.041
06	.036	.036	.029	.029	.023	.036	.043	.047	.051
08	.053	.057	.045	.045	.037	.052	.069	.085	.107
10	.074	.101	.059	.072	.056	.097	.103	.117	.152
12	.093	.132	.078	.094	.071	.136	.168	.180	.204
16	.190	.200	.165	.165	.114	.204	.243	.246	.266
20	.219	.262	.525	.222	.220	.281	.323	.356	.410
24	.397	.313	.640	.322	.330	.516	.497	.516	.730

NOTE: For definition of end fitting and connector, see paragraph 6.5.

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3.9.2 All connectors, space permitted, (see 6.5.3) shall be marked with the manufacturer's name or trademark, an appropriate material identification code and the dash size (mating tube) of the fitting, the manufacturer's part number, or standard part number (or as much of the part number as space permits) insuring that positive identification can be made. The package shall contain the complete part number.

3.10 Workmanship. Remove all burrs and break all sharp edges. Sealing surfaces shall be free of detrimental longitudinal and spiral tool marks. Unless a finer finish is specified on applicable drawings, sealing surfaces shall be smooth to a finish of 50Ra, except that annular tool marks up to 50Ra, as defined in ANSI B46.1 will be acceptable. All other machined surfaces shall be smooth to 125 Ra maximum and free of burrs and sharp edges. Unmachined surfaces, such as forging surfaces and bar stock flats shall be free of cracks, laps, and seams.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all parts of a given part number made from the same batch of material and processed at the same time.

4.2.1 Record maintenance. The supplier shall maintain a record of inspections applied to each lot.

4.2.2 Material certification. Records of chemical analysis and mechanical property tests showing conformance to the applicable material specifications shall be made available upon request.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

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4.4. First article inspection. The first article inspection shall consist of the inspections of this specification conducted in the order shown in Table III for the applicable assembly style and numbers.

4.4.1 Sampling for first article inspection. A random sample shall be selected consisting of test assemblies specified in Table III for each size, each fitting material and each tubing material for which inspection is desired. It is permissible to run each test on separate samples in the quantities indicated when specified or approved by the procuring activity. It is also permissible to combine certain tests in order to minimize sample quantities when approved by the procuring activity (see 6.2(e)). Threaded fittings designed for permanent attachment to adjacent tubing shall be assembled as fitting assemblies in accordance with the joint design feature (swage, braze or weld).

4.4.2 First article test report. The contractor shall prepare a report of the first article tests in accordance with MIL-STD-831 showing that the assemblies tested conform to this specification. Two copies shall be furnished to the procuring activity. Two additional copies shall be forwarded with the verification test samples. First article test report shall be in accordance with 6.2.2.

4.4.3 Verification test assemblies for the procuring activity. The contractor shall submit to the agency supplied in 6.2(g), together with the first article test report, five sample assemblies to be used as follows:

a. For a review of the mechanical construction of the product.

b. To perform any tests included in the specification, after reviewing the contractor's test report.

TABLE III. Inspection samples.

Assembly numbers	1-4	5-6	7-8	9-11	12-17	18-19	20-21
Assembly style (See Fig. 1)	C	A	A	B	A	C	A
L (See Fig. 1)	9"	9"	9"	6"	Per Table I ARP 1185		9"
Inspection sequence	4.6.1 4.6.2 4.6.8 4.6.5 4.6.3	4.6.1 4.6.2 4.6.10 4.6.7	4.6.1 4.6.2 4.6.10 4.6.4	4.6.1 4.6.2 4.6.9	4.6.1 4.6.2 4.6.6	4.6.1 4.6.2 4.6.6	4.6.1 4.6.2 4.6.11

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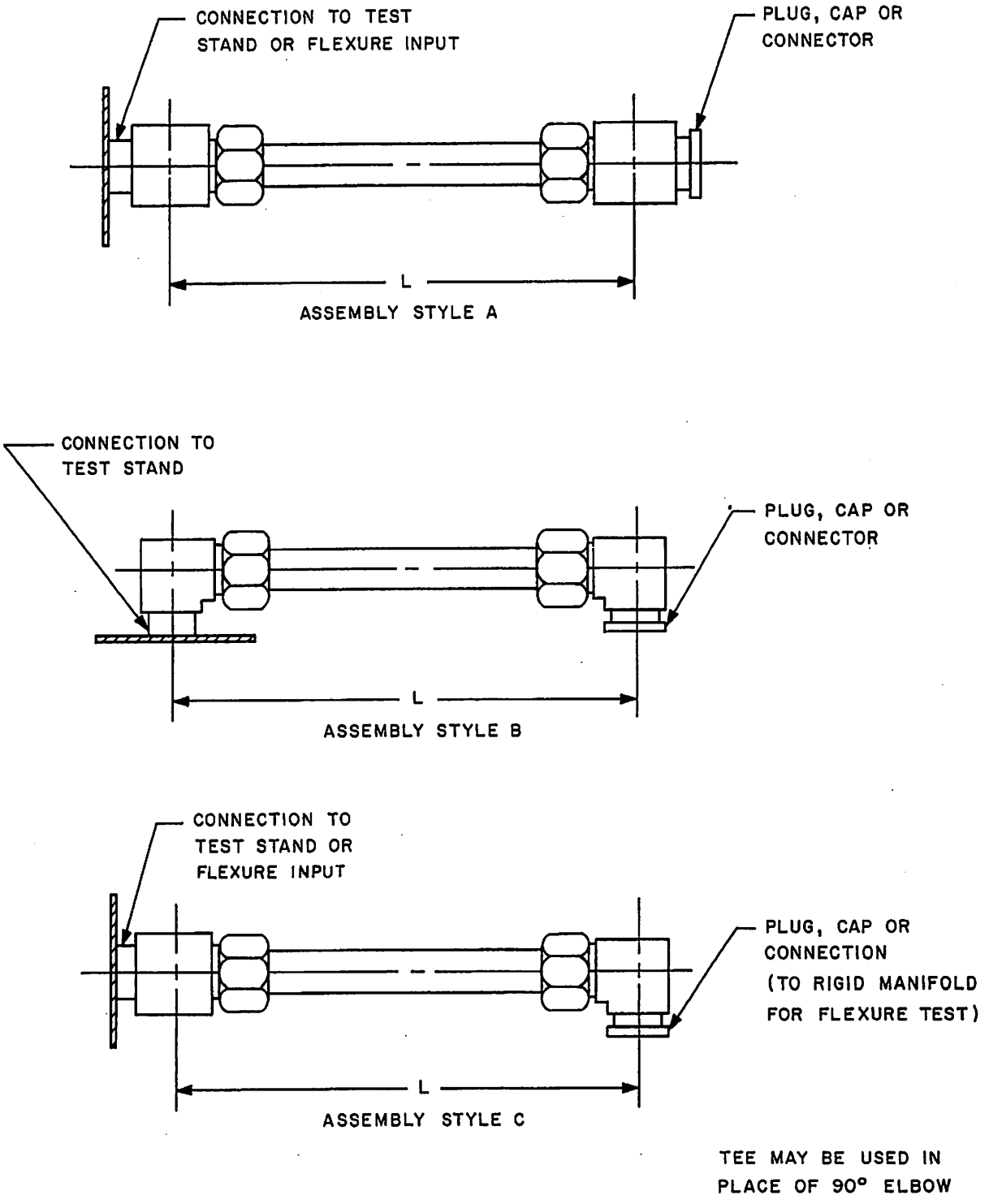


Figure 1. Assembly style

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4.5 Quality conformance inspection. A sample of each lot of fittings per MIL-STD-105, 1% AQL, Level II shall pass the examination of product when examined in accordance with 4.6.1.

4.5.1 Test conditions.

4.5.2 Test fluids. The test fluids used shall be suitable for, and compatible with, the fluid system temperature range and materials used.

4.5.3 Temperature. Except where otherwise specified, the tests shall be conducted at a room temperature of approximately 60° to 100°F and a fluid temperature of 60° to 170°F. Testing shall be conducted at maximum rated temperature ±10%, when required.

4.5.4 Torque. One end of each assembly (except flexure) shall be torqued to the minimum torque value and the other end torqued to the maximum torque value without application of any additional lubricant beyond that supplied by the manufacturer (see 6.5.6). If the boss seal is being tested, at least 50% of the samples will be torqued to the minimum torque and the balance to the maximum torque. Torque values for Titanium and CRES fittings shall be per Table IV.

TABLE IV. Torque requirements.

Fitting size	03	04	05	06	08	10	12	16	20	24
3000 psi										
Minimum (in-lbs)	60	120	120	180	360	492	600	840	1296	1500
Maximum (in-lbs)	108	168	192	300	480	660	840	1128	1680	1980
4000 psi										
Minimum (in-lbs)	84	144	156	240	420	576	720	1200		
Maximum (in-lbs)	108	168	192	300	480	660	840	1380		

4.6 Inspection methods.

4.6.1 Examination of product. All fitting assemblies and/or components shall be inspected for compliance with the workmanship, material, heat treat and marking requirements specified herein, and shall meet the dimensions and tolerances on the applicable inspection drawings.

4.6.1.1 All fittings and/or components shall be examined to ascertain that their preparation for delivery conforms to this specification.

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4.6.2 Proof pressure test. Test assemblies shall be connected to a source of pressure with one end free to move and proof pressure tested at a value equal to two times the nominal operating pressure for a minimum period of five minutes. Rate of pressure rise shall be 20,000 ± 5000 psi per minute. The test fluid may be water or suitable hydraulic fluid.

4.6.3 Burst pressure test. Test assemblies shall be connected to a source of pressure with one end free to move and burst tested at a value equal to four times the nominal operating pressure. The pressure shall be increased at a rate of 20,000 ± 5000 psi per minute until the assembly bursts or leaks. Two assemblies shall be tested at room temperature per 4.5.3 and two assemblies at maximum rated temperature.

4.6.4 Thermal shock test:

a. The test assembly shall be mounted in an environmental chamber, connected to a pressure source and completely filled with the specified test fluid. See Figure 2 for typical test setup.

b. The test assembly shall be pressurized to and maintained at the maximum system operating pressure and the ambient temperature raised to the maximum temperature requirement for the specified system and held at this temperature for a minimum of two hours. At the end of this period, while still at temperature, the hot test fluid shall be released and replaced with test fluid at the minimum temperature requirement of the specified system. Then, within 20 seconds, the fluid pressure shall be raised to 2 times the maximum system operating pressure and maintained at this pressure for one minute and then reduced to a maximum of 100 psi for an additional one-minute period. During the two one-minute pressurization periods, the assembly shall be checked for leakage or indication of malfunction. Restabilize the test system to room temperature.

c. The test assembly shall then be pressurized to and maintained at the maximum system operating pressure and the ambient chamber temperature lowered to the minimum temperature requirements of the specified system and held at this temperature for a minimum of two hours. At the end of this period while still at low temperature, the cold fluid shall be released and replaced with test fluid at the maximum temperature requirement of the specified system; and within 20 seconds, the fluid pressure shall be raised to 2 times the maximum system operating pressure. This pressure shall be maintained for one minute while checking the test assemblies for leakage or indication of malfunction.

d. Repeat testing per paragraphs b and c for a total of three times.

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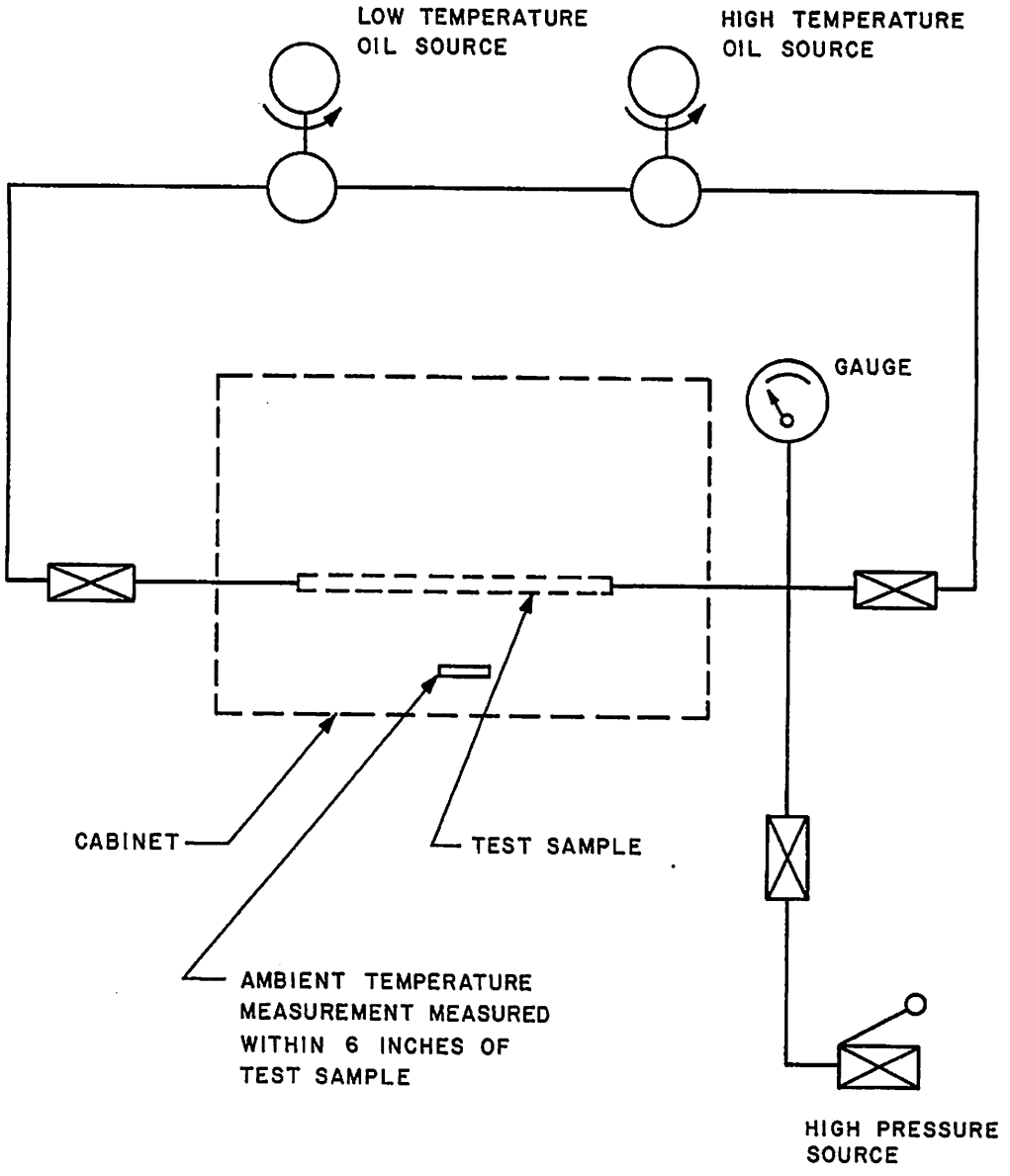


Figure 2. Typical setup for thermal shock testing

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4.6.5 Impulse test. Test assemblies except for 500 systems shall be tested in accordance with ARP 603 with the following deviations or additions.

a. Duration of test. 200,000 cycles minimum.

b. Temperature.

1. Fitting assemblies or components with non-metallic seals - maximum rated for 80,000 cycles; minimum rated (-65°F) for 40,000 cycles; and maximum rated for 80,000 cycles.

2. Fitting assemblies or components with metallic seals - maximum rated for 200,000 cycles.

NOTE: Test assembly must be stabilized at each temperature before beginning that portion of test.

c. Rate of rise. 175,000 to 300,000 psi/second.

d. Cycling rate. 70 ± 5 cycles per minute.

e. Wave form. Per ARP 603.

f. Surge Peaks. 143 to 157% of nominal operating pressure.

4.6.6 Flexure test. Test assemblies shall be tested in accordance with the flexure test procedure of ARP 1185 except as noted herein.

Test assemblies shall be tested to the minimum bending stress indicated in Table I at max rated temperature with static internal pressure equal to nominal operating pressure maintained. The bending stress shall be measured with suitable strain gages on each tube assembly.

NOTE: If it is desired to express the stress in terms of combined (pressure) plus bending stress, the pressure axial stress caused shall be calculated by formula:

$$\sigma_x = P \frac{D_i^2}{D_o^2 - D_i^2}$$

4.6.7 Tensile test. Test assemblies shall be mounted in a tensile test machine with no internal fluid or pressure. The tensile loading shall be increased at a head travel rate of .15 ± .10 in/min until failure is obtained.

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4.6.8 Repeated assembly. The test assemblies shall be tested with one end torqued to maximum torque and the other end torqued to minimum torque with no additional lubricant permitted for the first tightening. Each connection shall then be disassembled, with the sealing face then rotated 60 to 90 degrees, the fitting then lubricated with MIL-H-5606 hydraulic fluid on threads only and retorqued to the original torque a total of 25 times. After each fifth and the final tightening operation, the assembly shall be subjected to operating pressure for 5 minutes using air or nitrogen with the specimen submerged under water. No evidence of leakage shall be allowed. Every sixth cycle the connectors shall be interchanged.

No additional lubricant, other than MIL-H-5606 on the threads, is permitted at any point during test. The test fixture shall be such as to assure that all forces to join the assembly are induced into the joint assembly.

4.6.9 Stress corrosion test. The test assembly shall be installed in a test fixture which imposes a calculated stress level (using nominal tube dimensions) equal to 85 $\pm$ 5% of tubing yield strength (minimum per applicable tube specification) at the tubing - fitting assembly interface. The test assembly, pressurized to an internal pressure equal to the nominal system operating pressure, shall then be subjected to the standard salt spray test per FED-STD-151, Method 811.1 without removing the bending stress. After 50 hours exposure, the assembly shall be subjected to the burst pressure test at room temperature. The fittings shall then be cleaned and metallurgically examined, including sectioning of fitting and tubing.

4.6.10 Pneumatic test. The test assemblies shall be proof tested at room temperature at a value equal to two times the nominal operating pressure for a minimum period of five minutes. The test fluid shall be dry compressed air or nitrogen. The test assemblies shall be prepared without use of oil during assembly or prior to testing. Any oil or grease used for tubing attachment shall be completely removed before test.

4.6.11 Fire test. The assembly shall be mounted in a horizontal plane that lies on the axis of the burner.

The portion of the flame in which the assembly is to be tested shall deliver a minimum heat output of 4500 BTU per hour (1320 watts) as measured by a 1/2-inch (12.7 mm) copper tube mounted per AIR 1377. Width of the flame shall be such that the 2000°F (1100°C) temperature shall occur (as measured by thermocouple) for a minimum distance of 8 inches (203 mm). The entire fitting connection plus a length of tubing shall be subjected to the flame. During the test, the end connection subjected to the flame shall be subjected to inline vibration at the rate of 2000 cycles per minute through a total amplitude of not less

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than 1/8 inch (3.2 mm); i.e., a displacement of 1/16 inch (1.6 mm) each side of the neutral position. The assembly shall be subjected to SAE No. 20 oil at a pressure equal to the rated system pressure and at an oil temperature of 200-230°F (93-110°C). The flow rate shall be flow in gallons per minute equal to 1 x the ID^2 measured in inches.

5. PACKAGING

5.1 Preservation, packaging and packing. Fittings or their components, assembled if required shall be protected by capping all threaded ends and tube attachment ends with caps with closure material per MIL-C-5501 paragraph 3.2.1 or 3.2.2, prior to packaging. The parts shall then be skin-packed on 15" x 12" cardboard flats (or multiples thereof). Each flat shall be preprinted or a label affixed for identification. As an alternate, the parts may be packaged in nylon or polyethylene plastic bags with enclosed identification labels or the bags preprinted with identification. The packaged fittings shall be prepared for delivery in accordance with MIL-STD-794 for the applicable level as specified (see 6.2). Unit preservation, packaging and exterior packaging shall be level C, unless otherwise specified.

5.2 Marking of shipments. The interior labels (or preprinted flats or bags) and exterior shipping containers shall be marked in accordance with MIL-STD-129 and, if addition, shall include the following:

Manufacturer's name or trademark
Manufacturer's part number or Standard part number or FSCM
Customer's part number (if applicable)
Type and system pressure
Fitting material
Date of package

6. NOTES

6.1 Intended use. The fittings or components covered by this specification are intended for use in aerospace fluid systems under the conditions specified herein.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Fitting material (see 3.3.1)

c. Tubing material (see 3.3.2)

d. Type and system pressure (see 1.2)

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e. Manufacturer's, Military Standard Part Number or FSCM number, if available.

f. Sampling instructions for tests and when applicable, additional testing of fitting styles (such as tees, reducers, etc.) not included in the test samples (see 4.4.1).

g. Where the first article verification test samples (see 4.4.3) should be sent and instructions concerning the submittal of the test reports (see 4.4.2).

h. Applicable levels of preservation, packaging, and packing (see 5.1).

6.2.1 Procurement documents should specify the following provisions for first article inspection.

6.2.1.1 Whether first article or sampling inspection is required. When a contractor is in continuous production of the fitting from contract to contract, consideration should be given to waive the first article and/or sampling inspections. If first article inspection is required, indicate:

a. Where the first article inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

b. That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraphs of the Data Item Description (DID) Specification.

Paragraph	Data Requirements	Applicable DID
4.4.2	First Article Test Report	DI-T-5329
4.5	Quality Conformance Test Report	DI-T-5329

6.2.3 Sample test assemblies subjected to first article inspection shall not be considered, accepted or shipped as part of the contract or order.

6.3 Data. For the information of Contractors and Contracting Officers, any of the data specified in (a) applicable documents listed in Section 2 of this specification, or (b) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order.

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6.4 Material certification. Records of the chemical analysis and mechanical property tests showing conformance to the applicable specification shall be made available to the procuring activity's inspector upon request. Material heat treat and inspection records shall be retained for a period of five years from the date of raw stock procurement by manufacturer.

6.5 Definitions.

6.5.1 Fitting assembly. Fitting assembly as used herein is defined as the end fittings attached to the appropriate tubing with necessary connectors to conform to proper test sample (see 3.4.4 and Table III).

6.5.2 End fitting. End fitting as used herein is defined as the component or components which become permanently attached to the tubing end. This end fitting consists of either a nut (internally or externally threaded) with other components as necessary, or a single fitting with a threaded end.

6.5.3 Connector. Connector as used herein is defined as the replaceable component which connects end fittings or bosses or ports one with another by means of threads. Normally, these components are unions, boss adapters, 45 and 90 degree elbows, tees, and crosses.

6.5.4 Leakage. Leakage as used herein is defined as the escape of fluid (gaseous or liquid) from any point of the fitting assembly including the fitting-tube junction. This junction is defined to include one tube outside diameter in length beyond the fitting envelope. The envelope is the area from free parent tube material to free parent tube material or boss.

6.5.5 Failure. Failure as used herein is defined as any deformation that would contribute to seal leakage or would promote premature fatigue breakage, unless otherwise determined to be due to a tubing defect.

6.5.6 Torque. Torque as used herein is defined as any means of applying the necessary loads (or assembly stresses) to assure the sealing of the end fitting to the connector.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Preparing activity
Navy - AS
Project No. 4730-0144

Reviewer activities:
Army - AR, MI
Air Force - 11
DLA - CS

User activities:
Navy - OS
Army - GL, ME, AT, AR

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